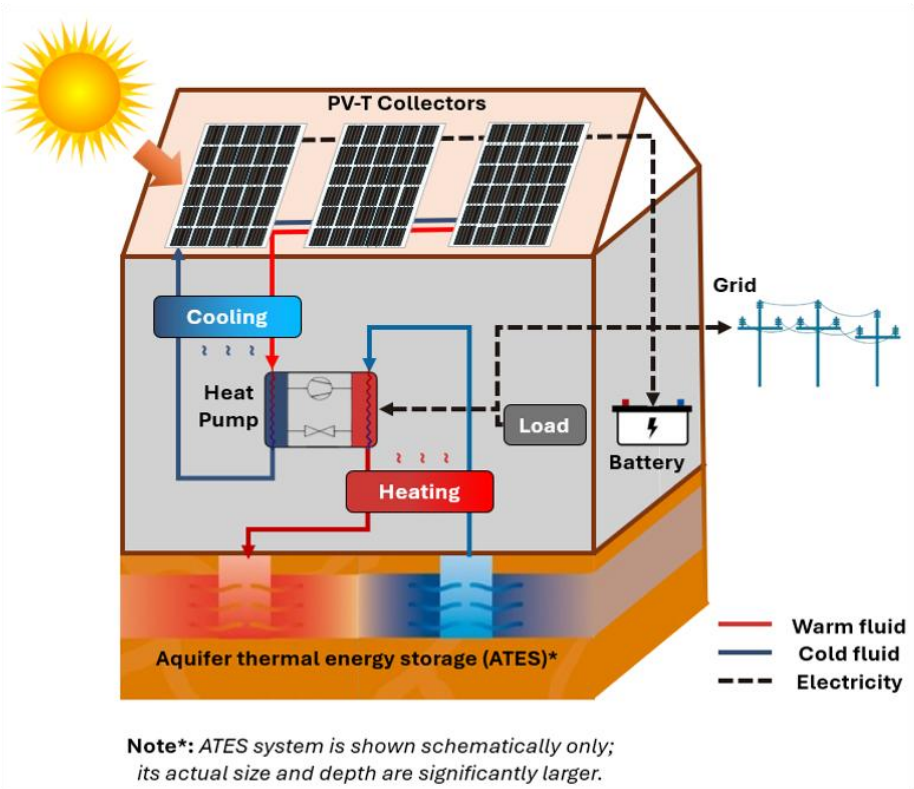


SOLSTICE V2

User Guideline

Step-by-step setup, operation and interpretation of the Excel-based model



Integrated SOLSTICE system schematic embedded in the current workbook^[1].

Solar Storage in Diverse Climates

SOLSTICE V2 workbook guidance - August 2026

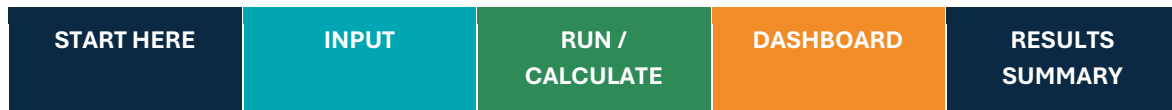
Purpose of this guide

Use this document when you first receive SOLSTICE V2, when you want to add a new location, or when you are unsure which sheet, input or result to use. It is written for normal model use; detailed component derivations remain in the thesis and scientific article.

[1] F. Lems, Z. Ul-Abdin, O. Isabella, R. Santbergen, Introducing SOLSTICE: A Modelling Tool for Assessing PV(T) Systems with Heat Pump and Seasonal Storage in Diverse Climates, manuscript submitted to Solar RRL, 2026.

0. Quick Start

The normal SOLSTICE workflow is intentionally short. Start with the home page, define the scenario, recalculate the model, and then inspect the visual and numerical outputs.



Before you begin

Open the .xlsm workbook in a version of Microsoft Excel that can run its VBA macro. If Excel asks whether to enable macros or active content, the HT-ATES update routine will only work when that functionality is available.

If you are using Berlin or Madrid

1. Open Start Here and choose the existing-location route.
2. Go to Input and select the location, number of households, results configuration and irradiance source.
3. Change only the intended user-input cells for building, solar collector and storage sizing.
4. Fully recalculate the workbook. For an HT-ATES case, run the ATES update when required.
5. Use Dashboard for the visual overview and Results_Summary for the numerical comparison.

If you are adding another location

1. Open data and add the location to the Location Registry.
2. Add the required 8,760-hour electricity-price, weather and solar data.
3. Add or select a 24-hour appliance and DHW demand profile.
4. Confirm that the data-completeness checks show OK.
5. Return to Input, select the new location and continue with the normal workflow.

Contents

Section	Topic
1	What SOLSTICE V2 is
2	Workbook structure and Start Here
3	Using an existing location
4	Adding a new location
5	Defining the scenario on Input
6	Solar irradiance and POA
7	Recalculation and HT-ATES update
8	Dashboard and Results Summary
9	Configurations and operating modes
10	Advanced settings and calculation sheets
11	How to interpret the main results
12	Modelling scope and troubleshooting

1. What SOLSTICE V2 Is

SOLSTICE V2 is an hourly Excel-based framework for transparent, early-stage assessment of integrated residential energy systems. It combines solar generation, heat-pump operation, short-term electrical storage, seasonal thermal storage, residential demand, electricity prices and grid exchange in one workbook.

The model is intended to help users compare system concepts and understand the interactions between the electrical and thermal subsystems without requiring specialist simulation software or programming knowledge.

Main technologies represented

Component	Role in SOLSTICE V2
PV	Photovoltaic modules generate electricity from plane-of-array solar irradiance.
PVT	Photovoltaic-thermal collectors generate electricity and provide a fluid-loop heat source for the heat pump.
HP	The heat pump supplies space heating, domestic hot water, cooling and - where applicable - HT-ATES charging.
EB	The electrical battery shifts surplus solar electricity to later hours and increases local solar use.
HT-ATES	High-temperature aquifer thermal energy storage shifts useful heat seasonally from solar-rich periods to heat-demand periods.
Grid	The electricity grid acts as the final source or sink for electricity when local generation and storage are insufficient or excessive.

What the model is for

- Comparing system configurations on a common hourly basis.
- Exploring the effect of climate, demand, collector sizing and storage sizing.
- Evaluating annual and peak grid exchange, solar self-use, self-sufficiency, HP electricity and COP, operational electricity cost and HT-ATES behaviour.
- Understanding how short-term electrical storage and seasonal thermal storage complement each other.

What the model is not for

- Detailed engineering design of a heat pump, battery, solar collector or aquifer installation.
- Site-specific subsurface design or groundwater-flow simulation.
- Sub-hourly control design, forecast-based dispatch or full techno-economic optimisation.

Key principle

SOLSTICE V2 should be read as a transparent system-screening model. A result can indicate whether a concept or sizing direction is promising, but detailed engineering design requires additional component validation and site-specific modelling.

2. Workbook Structure and Start Here

The current workbook contains a user-facing layer, supporting input/calculation sheets and advanced backend sheets. Normal users should spend most of their time on Start Here, Input, Dashboard and Results_Summary.

Start Here

Start Here is the home page. It explains what SOLSTICE is, offers two entry routes, provides a workbook guide and shows the recommended navigation sequence.

Route	Use when	Path
Option A - Use an existing location	Berlin or Madrid already covers the scenario you want to study.	Input -> Run / Calculate -> Dashboard -> Results Summary
Option B - Add a new location	You need another city, climate, demand profile or country-specific price dataset.	data -> Input -> Run / Calculate -> Dashboard -> Results Summary

Worksheet map

Sheet	Purpose	Normal user action
Start Here	Home page, recommended workflow and navigation.	Start here.
Input	Scenario setup, location selection and main building/solar/storage inputs.	Primary user sheet.
data	Location registry, 8,760-hour data, demand profiles and completeness checks.	Use when reviewing or adding location data.
Dashboard	Visual overview of system performance.	Review results visually.
Results_Summary	Annual, peak, cost, COP, ATES and validation metrics.	Review numerical results.
Advanced_Settings	Detailed component assumptions and technical parameters.	Change only when justified.
POA_Calc	Plane-of-array irradiance calculation and source selection.	Normally inspect rather than edit.
HP / HP+PV / HP+PV+EB / ...	Hourly calculations for each system configuration.	Do not edit during normal use.
ATES_model	HT-ATES calculations and hourly storage update.	Advanced calculation sheet.
HP_Modes_Template	Reference/backend logic for heat-pump modes.	Backend/template sheet.

Naming note

Several calculation-sheet names still use “TB”. In the current SOLSTICE V2 model, this refers to the seasonal thermal-storage branch represented by HT-ATES.

3. Using an Existing Location

Berlin and Madrid are currently registered in the workbook. This is the fastest route for first-time use and for standard scenario studies.

Step-by-step

1. Open Input from Start Here.
2. In Active Scenario Setup, select the location. The location-dependent demand, weather and price information updates automatically.
3. Enter the number of households. Use a positive whole number.
4. Select the Results configuration you want to inspect in detail.
5. Select the irradiance source: the SOLSTICE POA calculator or an imported PVGIS POA series when one is available.
6. Adjust the intended building, collector and storage sizes.
7. Check that the Active data status shows OK.
8. Fully recalculate the workbook. For a configuration containing HT-ATES, run the ATES update when required.
9. Open Dashboard to review the system visually and Results_Summary to compare numerical indicators.

Current location-dependent inputs

Input type	Examples loaded with the selected location
Location metadata	Country, simulation year and active demand profile.
Demand	Annual appliance electricity demand, annual DHW demand and 24-hour profile shape.
Weather	Ambient temperature and wind speed.
Solar resource	GHI, DNI, DHI, solar azimuth, solar zenith and optional PVGIS POA.
Electricity prices	Hourly wholesale price series, target household purchase price and export price.

Do not overwrite linked cells

The Input sheet already distinguishes user inputs from automatically selected or calculated values. Avoid typing over grey/linked outputs simply to force a result; change the corresponding user input or location data instead.

4. Adding a New Location

Use the data sheet when a city is not already available. The workbook is structured around a Location Registry, a 24-hour demand-profile table and a location-specific hourly table. The current completeness checks expect 8,760 hourly records, corresponding to a non-leap simulation year.

SOLSTICE — LOCATION AND HOURLY INPUT DATA			
ACTIVE LOCATION SETUP		LOCATION SELECTOR SUPPORT	DATA COMPLETENESS CHECKS
Setting / Parameter	Selection / Value	Unit	
Simulation year	2023	year	
Selected location	Berlin	location	
Demand profile	Germany (BDEW H25)	profile	
Target buy price	0.39	€/kWh	
Export price	0.08	€/kWh	
Annual appliance demand	1827.0	kWh/year	
Annual DHW demand	3051.4	kWh/year	
Daily appliance demand	5.0	kWh/day	
Daily DHW demand	8.4	kWh/day	
TARGET PURCHASE PRICE COMPATIBILITY		Location	Target price (€/kWh)
Country			
Germany	Berlin		03.39
Spain	Madrid		03.29
LOCATION SELECTOR SUPPORT		Location	Demand profile
Named range		input_Location_List	
		Berlin	Germany (BDEW H25)
		Madrid	Spain (IDAE)
Check	Count / Value	Required	Status
Location exists	1/1		OK
Demand profile exists	24/24		OK
Appliance demand > 0	1827.0 > 0		OK
DHW demand > 0	3051.4 > 0		OK
Wholesale-price rec	8760/8760		OK
Temperature records	8760/8760		OK
Wind-speed records	8760/8760		OK
GHI records	8760/8760		OK
DHI records	8760/8760		OK
DNI records	8760/8760		OK
Solar-azimuth records	8760/8760		OK
PVGIS POA records	0/Optional unless		OK
Appliance fraction < 1	100.0% < 1		OK
DHW fraction total	100.0% < 1		OK
Overall data status	OK	All required checked	OK
LOCATION REGISTRY		Location	Country
		Berlin	Germany
		Madrid	Spain
ACTIVE ELECTRICITY PRICE DATA		Hour	Wholesale purchase price (€/MWh)
		1	-15.41
		2	-15.70
		3	-17.51
		4	-17.21
		5	-13.82
		6	-12.28
		7	-3.65
		8	-11.29
		9	-22.7
		10	-0.33
		11	-0.31
		12	-4.10
		13	5.68
ACTIVE WEATHER DATA		Hour	Ambient temperature (°C)
		1	11.5
		2	14.9
		3	14.7
		4	14.4
		5	14.2
		6	13.9
		7	14.0
		8	14.3
		9	14.1
		10	14.2
		11	14.5
		12	14.7
		13	14.8
ACTIVE DEMAND PROFILE		Hour	Appliance fraction
		1	0.0%
		2	2.7%
		3	2.5%
		4	2.5%
		5	2.0%
		6	3.0%
		7	3.8%
		8	4.1%
		9	3.0%
		10	3.7%
		11	3.8%
		12	4.1%
		13	4.3%
ACTIVE SOLAR AND POA DATA		Hour	GHI (W/m²)
		1	0.0
		2	0.0
		3	0.0
		4	0.0
		5	0.0
		6	0.0
		7	0.0
		8	9.3
		9	52.4
		10	94.5
		11	119.6
		12	102.9
		13	125.2

Figure 1. Data sheet structure: active location setup, completeness checks, registry and hourly input tables (from the thesis workbook appendix).

4.1 Add the location to the Location Registry

Field	What to enter
Location	Unique city/location name used by the selector.
Country	Country associated with prices and reporting.
Demand profile	Name of the 24-hour profile to use.
Simulation year	Year represented by the hourly dataset.
Appliance demand	Annual household appliance electricity demand in kWh/year.
DHW demand	Annual domestic-hot-water thermal demand in kWh/year.
Target buy price	Desired annual-average household purchase price in €/kWh.

Field	What to enter
Export price	Country/location-specific export remuneration in €/kWh.

4.2 Add the 8,760-hour location dataset

- Wholesale electricity price
- Ambient temperature
- Wind speed
- Global horizontal irradiance (GHI)
- Direct normal irradiance (DNI)
- Diffuse horizontal irradiance (DHI)
- Solar azimuth
- Solar zenith
- PVGIS plane-of-array irradiance, only when the PVGIS route will be used

Price-series logic

SOLSTICE preserves the hourly variation of the wholesale electricity-price series and uses the target purchase price to correct the annual average toward the selected household price level. The export price is handled separately.

4.3 Add or select a 24-hour demand profile

Each demand profile contains 24 hourly appliance fractions and 24 hourly DHW fractions. For each daily profile, the fractions should sum to 1.0 (100%). The annual demand from the Location Registry scales the selected shape to the appropriate absolute demand.

4.4 Use the completeness checks before continuing

Check	Expected condition
Location exists	The selected location must be present exactly once in the registry.
Demand profile exists	The profile should contain 24 records.
Annual demands available	Appliance and DHW annual demand must be greater than zero.
Hourly records	Price, temperature, wind and required solar variables should each contain 8,760 records.
PVGIS POA	Required only when PVGIS is selected as the irradiance source.
Profile totals	Appliance and DHW fraction totals should each equal 1.0.
Overall data status	Proceed when the sheet reports OK for the required checks.

If a new location does not appear on Input

First check the Location Registry name, then confirm the data-completeness checks. A location should not be treated as ready until the required data rows and demand-profile records are complete.

5. Defining the Scenario on Input

Input is the main user interface for a simulation. The current sheet explicitly instructs the user to change the blue dropdown selections and the intended numerical input cells, while linked or calculated values update automatically.



Figure 2. Input sheet showing active scenario, location/demand data, building inputs, solar sizing, storage sizing and supporting plots (from the thesis workbook appendix).

5.1 Active Scenario Setup

Parameter	What it controls
Location	Which registered city dataset is active.
Number of households	Scale used for collective/system quantities.
Results configuration	Configuration selected for detailed active-scenario inspection.
Irradiance source	Hourly POA series used by the solar components.

5.2 Building and comfort inputs

- Building heat-loss coefficient [W/K] - controls heat transfer to/from ambient.
- Effective thermal mass [J/K] - controls the building temperature response.
- Effective sun-facing window area [m²] and solar-gain factor - control passive solar gains.
- Minimum and maximum indoor temperature [°C] - define the heating/cooling comfort band.

5.3 Solar orientation and sizing

- Module tilt [degrees].
- Module azimuth [degrees]; the workbook convention uses 180° for south.
- Ground albedo [-].
- PV area [m²].
- PVT area [m²].
- Separate PV and solar-thermal collector areas for the ST configuration.

5.4 Storage and HT-ATES sizing

Input	Meaning
Electrical-battery capacity [kWh]	Usable short-term electrical storage per model setup.
HT-ATES hot-well volume [m³]	Reference/effective hot-side storage volume.
HT-ATES cold-well volume [m³]	Reference/effective cold-side storage volume.
Annual injected water volume [m³/year]	Annual volume used by storage/recovery screening.
Maximum ATEs flow rate [m³/h]	Upper hourly equivalent charging/discharging flow constraint.

Recommended approach

Change one design choice at a time when performing a sensitivity study. This makes it easier to identify which parameter caused the change in grid exchange, HP operation, storage behaviour or cost.

6. Solar Irradiance and the POA Calculator

All solar components use plane-of-array (POA) irradiance: the solar irradiance incident on the selected collector plane rather than horizontal irradiance alone.

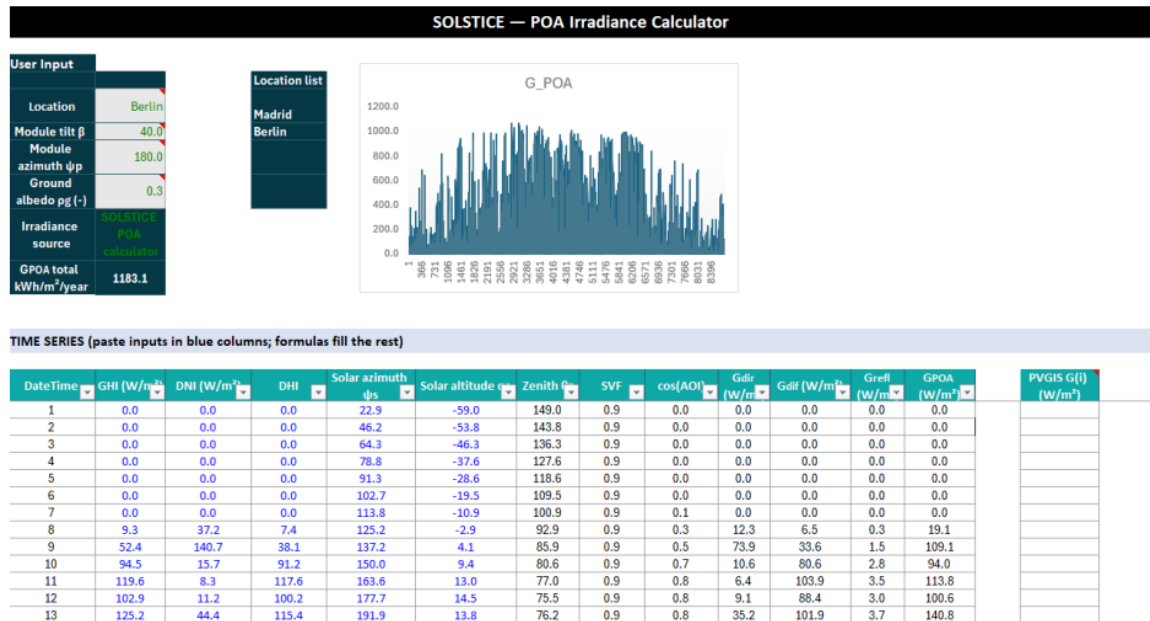


Figure 3. POA_Calc sheet with orientation inputs, hourly solar-resource data, calculated POA and optional PVGIS POA input (from the thesis workbook appendix).

Two irradiance routes

Route	Required data	When to use
SOLSTICE POA calculator	GHI, DNI, DHI, solar azimuth and solar zenith, plus module tilt, azimuth and albedo.	Use when you want the workbook to calculate POA internally.
PVGIS POA	A complete hourly PVGIS POA series for the selected orientation.	Use when you already have external POA data and want to bypass the internal transposition result.

What to check

- The selected irradiance source on Input is the one you intend to use.
- Annual irradiation used is plausible for the location and orientation.
- If PVGIS is selected, the PVGIS POA column contains all required hourly values.
- If the internal calculator is selected, the underlying GHI/DNI/DHI and solar-position data are complete.

One common solar input

The selected hourly POA series is the common irradiance input used by the PV, PVT and solar-thermal calculations. This avoids using different solar-resource assumptions across components.

7. Recalculation and the HT-ATES Update

Most workbook formulas update through Excel calculation. HT-ATES results also use a VBA update routine, so storage results must be refreshed after relevant changes.

When to recalculate and run the ATES update

- After changing the active location.
- After changing solar collector area.
- After changing the HT-ATES volumes, injection volume, flow rate or related advanced ATES parameters.
- Before using HT-ATES diagnostics or the Dashboard ATES KPIs for a changed scenario.

Recommended sequence

1. Make the input changes.
2. Force a full workbook recalculation in Excel.
3. For an HT-ATES case, click the RUN ATES / Update ATES Results control in the workbook.
4. Open Results_Summary and check the ATES snapshot status and cyclic status.
5. Then interpret Dashboard and Results_Summary.

Important

Results_Summary contains an explicit warning: after changing the location, ATES inputs or solar collector area, fully recalculate the workbook and run the Update ATES Results macro before using the ATES results.

How to recognise a current result

For the active ATES configuration, the validation area in Results_Summary should indicate that the ATES snapshot is current. “NOT ACTIVE” is expected for configurations that are not the currently updated ATES case.

If the ATES result looks wrong

- Check the active ATES results configuration.
- Check that the location data status is OK.
- Check the storage volume, initial/operating temperatures and maximum flow rate.
- Recalculate and rerun the ATES update.
- Inspect the ATES diagnostics for depletion, accumulation or remaining unmet thermal demand.

8. Dashboard and Results Summary

8.1 Dashboard - visual overview

The Dashboard is the fastest way to understand what the selected scenario is doing. It combines headline KPIs with configuration comparisons and charts for grid exchange, cost, self-use, self-sufficiency, HP performance and HT-ATES behaviour.



Figure 4. Dashboard visual overview (from the thesis workbook appendix).

Dashboard area	What it tells you
Configuration comparison	How generation, consumption, autonomy and cost differ between concepts.
Grid import / export	Annual and peak interaction with the electricity grid.
Net electricity cost	Operational import cost after export revenue.
Self-use & sufficiency	How much solar is used locally and how much electricity demand is covered without grid import.
Heat pump & COP	Annual HP electricity and weighted performance by duty.
ATES performance	Charging, useful heat supply, inventory and cyclic balance.

8.2 Results_Summary - numerical comparison

Use Results_Summary when you need the exact annual, peak, COP, cost and HT-ATES values rather than a graphical overview.

SOLSTICE V2 — Results Summary

Note: After changing the location, ATEs inputs, or solar collector area, fully recalculate the workbook and run the Update ATEs Results macro before using the ATEs results.

SECTION 1 — RUN INFORMATION

Active location

Berlin

Linked to existing selector / source

Simulation year

2023

Derived from input time series

Number of household

1000.0

Linked to HT-ATES input

Active weather dataset

Table4(Amb)

Formula source used for weather rows

Electricity price coun

Germany

Mapped from active location

Datetime last review

#####

Updates on recalculation

ATES cyclic-balance

1.0%

Editable screening tolerance, not a physical model parameter

SECTION 2 — CONFIGURATION COMPARISON

Configuration	Solar collector type	Total solar collector area [m²]	Electrical battery capacity [kWh]	Annual solar electricity generation [kWh]	Annual total electricity consumption [kWh]	Heat-pump electricity consumption [kWh]	Annual grid export [kWh]	Annual grid import [kWh]	Peak hourly grid export [kW]	Peak hourly grid import [kW]	Solar self-consumption [%]	Electric self-sufficiency [%]	Weighted COP for space heating [-]	Weighted COP for domestic hot water	Weighted COP for ATEs charging [-]	Average import price [€/kWh]	Annual import cost [€]	Annual export revenue [€]	Net elect
HP	PV	25.0	-	0.0	4893.5	3066.5	4893.5	0.0	13	0.0	0.0%	0.0%	3.2	2.9	-	0.39	1501.70	0.00	
HP+PV	PV	25.0	-	5730.8	4893.5	3066.5	304.3	3852.1	1.7	4.2	32.8%	38.4%	3.2	2.9	-	0.40	1500.13	623.58	
HP+PV+EB	PV	25.0	10.0	5730.8	4893.5	3066.5	208.0	2746.6	1.7	4.2	52.1%	58.7%	3.2	2.9	-	0.40	1502.20	620.63	
HP+PV+EB+TB	PV	25.0	10.0	5730.8	5332.8	4305.6	460.0	107.8	0.3	3.5	97.6%	92.4%	3.3	2.9	2.2	0.39	1476.23	115.64	
HP+PV+EB+TB	PVT	25.0	10.0	5827.8	5250.8	3423.4	428.4	891.2	0.3	3.7	84.7%	81.8%	3.7	3.8	2.8	0.38	1467.83	114.06	
PV+EB+ST+TB	PV+ST	30.0	10.0	3438.5	3541.5	0.0	251.0	944.4	0.3	2.8	52.2%	07.25%	-	-	-	0.39	137.24	1138.05	

Notes: Net elect is electricity export results for PV+EB+ST+TB result presented with the negative electric heating sign to indicate ATEs is fuelled.

SECTION 3 — ATEs DIAGNOSTICS

Configuration	ATEs recovery efficiency [%]	Initial useful thermal inventory [kWh]	Actual annual heat charged [kWh]	Recoverable annual heat [kWh]	Useful heat supplied from ATEs [kWh]	Final useful thermal inventory [kWh]	Cyclic-balance difference [%]	Cyclic-balance difference [kWh]	Minimum useful inventory [kWh]	Remaining useful thermal demand [kWh]	Cyclic-status
HP+PV+EB+TB	83.3%	2313.3	5945.8	4866.8	4866.8	2313.2	11.8	0.5%	27.8	4638.7	0.0 PASS
HP+PV+EB+TB	83.3%	2313.3	5945.8	4866.8	4866.8	2313.2	11.8	0.5%	27.8	4638.7	0.0 PASS
PV+EB+ST+TB	83.3%	2313.3	5945.8	4866.8	4866.8	2313.2	11.8	0.5%	27.8	4638.7	0.0 PASS

SECTION 4 — VALIDATION CHECKS

Configuration	Electrical energy-balance error [kWh]	Battery minimum state of charge [kWh]	Battery maximum state of charge [kWh]	Hours battery SOC < 0	Hours battery SOC > 100	PVT converg error	ATEs snapshot status
HP	0.0	-	-	-	-	-	NOT ACTIVE
HP+PV	0.0	-	-	-	-	-	NOT ACTIVE
HP+PV+EB	0.0	0.0	10.0	0.0	0.0	-	NOT ACTIVE
HP+PV+EB+TB	0.0	0.0	10.0	0.0	0.0	-	CURRENT
HP+PV+EB+TB	0.0	0.0	10.0	0.0	0.0	-	NOT ACTIVE
PV+EB+ST+TB	0.0	0.0	10.0	0.0	0.0	-	NOT ACTIVE
Number of formula-err	0.0						

SECTION 5 — JULY / DECEMBER GRID IMPORT / EXPORT [kWh]

Configuration	July Import [kWh]	July Export [kWh]	December Import [kWh]	December Export [kWh]
HP	0.0	0.0	0.0	0.0
HP+PV	0.0	0.0	0.0	0.0
HP+PV+EB	0.0	0.0	0.0	0.0
HP+PV+EB+TB	0.0	0.0	0.0	0.0
HP+PV+EB+TB	0.0	0.0	0.0	0.0
PV+EB+ST+TB	0.0	0.0	0.0	0.0

Figure 5. Results_Summary with configuration comparison, HT-ATES diagnostics and validation checks (from the thesis workbook appendix).

Provisional configuration

The current Results_Summary flags the PV+EB+ST+TB electricity-consumption, grid-import and electricity-cost results as provisional until its auxiliary electric-heating logic is finalised. Treat this alternative configuration separately from the five primary comparison cases.

9. Configurations and Operating Modes

9.1 Primary system configurations

Configuration	Concept
HP	Reference case. The heat pump supplies thermal demand using grid electricity.
HP + PV	PV supplies part of household and HP electricity; remaining surplus is exported.
HP + PV + EB	The battery stores short-term PV surplus before grid export.
HP + PV + EB + HT-ATES	Surplus PV electricity can operate the HP to charge seasonal thermal storage; useful stored heat can later supply heating/DHW.
HP + PVT + EB + HT-ATES	PVT replaces PV and supplies both electricity and a fluid-loop heat source to the HP.
PV + EB + ST + TB	Alternative solar-thermal branch retained in the workbook; currently flagged as provisional for some electricity/cost outputs.

9.2 Operating modes are energy-flow pathways

The modes are not mutually exclusive system states. More than one pathway can be active in the same hourly time step.

Mode	Meaning
Mode I - Household appliance electricity supply	PV/PVT is used first, then the battery, then the grid for the remaining appliance demand.
Mode II - Electricity-surplus operation	Solar surplus first charges the battery; additional surplus can charge HT-ATES via the HP; unused electricity is exported.
Mode III - Electricity-shortage and auxiliary HP operation	Battery electricity is used before grid import. If HT-ATES cannot cover the thermal demand, the HP supplies the remainder.
Mode IV - PVT-assisted HT-ATES charging	In the PVT configuration, surplus PVT electricity operates the HP while the PVT fluid loop supplies source heat; the upgraded heat charges HT-ATES.
Mode V - Direct HT-ATES heat supply	When useful stored heat is available, HT-ATES supplies space heating and DHW directly without HP operation; any remainder passes to auxiliary HP supply.

Energy priority

A useful mental model is: direct local use first, short-term electrical storage second, seasonal thermal storage when surplus is available, and the grid as final electrical backup or export sink.

10. Advanced Settings and Calculation Sheets

Advanced_Settings contains technical component assumptions. The values define component behaviour across all configurations, so they should not be changed casually.

Component	Advanced parameters
PV / PVT	Reference efficiency, temperature coefficient, absorptance, heat-loss coefficient and PVT internal heat-transfer parameters.
Heat pump	Minimum temperature lift, Carnot fractions, duty-specific sink temperatures and air-/water-source offsets.
Electrical battery	Round-trip efficiency.
HT-ATES	Operating temperatures, useful cutoff temperature, maximum flow rate, well volumes, aquifer geometry, hydraulic properties and recovery-efficiency screening inputs.

Rule for advanced changes

Only change an Advanced_Settings value when you have component data, literature support, a calibration objective or a clearly documented sensitivity-analysis reason. Record the changed assumption when comparing results.

Hourly configuration sheets

Each configuration has a dedicated hourly worksheet. These sheets expose the full energy-flow calculations and are useful for debugging or detailed analysis, but they are not normal user-input sheets.

[illegible]

Figure 6. Example configuration-level hourly calculation sheet (from the thesis workbook appendix).

When to inspect them

- Tracing why a particular hour imports or exports electricity.
- Checking battery charging/discharging and state of charge.
- Inspecting HP electricity, COP and thermal-duty allocation.
- Following HT-ATES charging/discharging and remaining useful inventory.
- Debugging an unexpected Dashboard or Results Summary value.

11. How to Interpret the Main Results

The Results_Summary sheet groups the main indicators into configuration performance, HT-ATES diagnostics and validation checks. Use several indicators together rather than judging a configuration by one number alone.

Cyclicity matters

A configuration can look attractive in one year while gradually emptying or overfilling the seasonal store. Always check the cyclic-balance difference and status when comparing HT-ATES designs.

Indicator	Interpretation
Annual solar electricity generation [kWh]	Total PV/PVT electricity produced over the year.
Annual total electricity consumption [kWh]	Total modeled electrical consumption, including household and applicable system electricity.
Annual HP electricity consumption [kWh]	Electricity used specifically by the heat pump.
Annual grid import / export [kWh]	Total electricity taken from or delivered to the grid.
Peak hourly grid import / export [kW]	Largest hourly exchange level; useful for assessing grid stress.
Solar self-consumption [%]	Share of generated solar electricity used locally rather than exported.
Electrical self-sufficiency [%]	Share of electricity demand covered without grid import.
Weighted COP - SH / DHW / ATEs [-]	Energy-weighted heat-pump performance for each thermal duty during active operation.
Average import price [€/kWh]	Average price paid for imported electricity in the modeled scenario.
Annual import cost [€]	Cost of grid electricity imports.
Annual export revenue [€]	Revenue from electricity export.
Net annual electricity cost [€]	Import cost minus export revenue; operational electricity only.

HT-ATES diagnostics

ATES metric	Interpretation
Recovery efficiency	Scenario-level fraction of charged thermal energy treated as recoverable useful inventory.
Initial / final useful inventory	Useful thermal energy above the minimum useful temperature at the start/end of the year.
Annual heat charged	Thermal energy actually sent to storage.
Recoverable charged heat	Charged heat after applying the recovery-efficiency constraint.
Useful heat supplied from ATEs	Heat actually delivered from storage to residential thermal demand.
Cyclic-balance difference	Difference between final and initial useful inventory.
Cyclic status	PASS when the end state is sufficiently close to the initial state; otherwise indicates depletion or accumulation.
Remaining unmet thermal demand	Thermal demand that the modeled storage/auxiliary pathway did not satisfy under the applicable logic.

12. Modelling Scope, Troubleshooting and References

12.1 Main modelling assumptions to remember

- System operation is evaluated with hourly energy balances; sub-hourly transients, start-up/shutdown and component response times are not resolved.
- Energy flows follow predefined rule-based priorities; forecast-based or cost-optimal dispatch is not applied.
- Heat exchangers, connecting hydraulic loops and auxiliary pumping are not modelled in full detail.
- The electricity grid is assumed to remain available without an explicit import or export capacity limit.
- For collective HT-ATES cases, households are treated as identical and synchronised, and collective quantities are scaled from the single-household profile.
- The HT-ATES representation is a system-level/lumped storage model rather than a spatial groundwater-flow and heat-transport model.

12.2 Troubleshooting

Problem	First check
New location is missing	Check the Location Registry name, the demand-profile entry and overall data status.
Data status says CHECK	Use the data completeness table to identify the missing 8,760-hour series or profile total.
Solar output looks wrong	Verify the irradiance source, orientation inputs and annual irradiation used.
PVGIS selected but no result	Confirm that all required hourly PVGIS POA records are populated for the location.
Dashboard values look stale	Fully recalculate the workbook; if HT-ATES is involved, rerun the ATES update.
ATES snapshot is not current	Select/update the intended ATES configuration, recalculate and run the ATES update routine.
Storage depletes or accumulates	Check collector area, storage size, recovery efficiency, charging temperature, maximum flow rate and demand balance.
Need to understand a single hour	Open the corresponding configuration calculation sheet and trace the hourly energy-flow columns.
PV+EB+ST+TB cost result is being used	Check the provisional-results warning in Results_Summary before drawing conclusions.

Recommended distribution

Keep the workbook and documentation together. If links are hosted online, the .xlsm file can be distributed by itself. If files are distributed locally, keep them in a consistent Documentation folder so hyperlinks remain easy to maintain.

End of SOLSTICE V2 User Guideline

For component equations, validation and scientific interpretation, consult the thesis and scientific article.